

✓ Lab11: Artificial Neural Network (ANN)

An Artificial Neural Network (ANN) is a machine learning model inspired by how the human brain works.

Just like your brain has neurons that pass signals to each other, an ANN has artificial neurons that take input, process it, and produce an output.

Objective

Learn how a simple **Artificial Neural Network (ANN)** can predict exam scores from **study hours**.

We'll keep this super simple: tiny data, a tiny model, and clear steps.

✓ Step 1: Import the Needed Tools

These are the “ingredients” for our program.

- `numpy` for numbers
- `matplotlib` for graphs
- `Sequential` and `Dense` to build a small ANN

```
#!pip install tensorflow
```

```
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
```

✓ Step 2: Make a Small Dataset

We give the computer examples of **study hours** and **exam scores** so it can learn the pattern.

```
# Study hours (input) and exam scores (output)
study_hours = np.array([1, 2, 3, 4, 5, 6, 7, 8], dtype=float)
exam_scores = np.array([35, 40, 50, 55, 65, 70, 80, 90], dtype=float)

# Quick check
print("Study Hours:", study_hours)
print("Exam Scores:", exam_scores)
```

```
Study Hours: [1. 2. 3. 4. 5. 6. 7. 8.]
Exam Scores: [35. 40. 50. 55. 65. 70. 80. 90.]
```

✓ Step 3: Build the ANN Model

- **Input layer:** 1 value (Study Hours)
- **Hidden layer:** 64 neurons with **ReLU** activation (keeps useful signals, ignores negatives)
- **Output layer:** 1 value (Predicted Exam Score) with **linear** activation

```
# Build a simple ANN
model = Sequential()

# Input + Hidden layer (64 neurons). input_dim=1 → one input feature: Study Hours
model.add(Dense(64, input_dim=1, activation='relu'))

# Output layer (1 neuron) → predicts the exam score
model.add(Dense(1, activation='linear'))

# Compile: tell the model how to learn
# optimizer='adam' is like a helpful teacher
```

```
# loss='mean_squared_error' measures how far off predictions are
model.compile(optimizer='adam', loss='mean_squared_error')

# Show the model summary
model.summary()
```

```
c:\Users\UNS\anaconda3\Lib\site-packages\keras\src\layers\core\dense.py:95: UserWarning: Do not pass an `input_shape` to
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 64)	128
dense_1 (Dense)	(None, 1)	65

Total params: 193 (772.00 B)
Trainable params: 193 (772.00 B)
Non-trainable params: 0 (0.00 B)

✓ Step 4: Train the Model (Let It Learn!)

The model practices for **200 epochs** (200 rounds). More practice → better learning (usually).

```
history = model.fit(study_hours, exam_scores, epochs=800, verbose=1)

# Plot training loss to see learning progress
plt.plot(history.history['loss'])
plt.title('Training Loss Over Time')
plt.xlabel('Epoch')
plt.ylabel('Loss (MSE)')
plt.show()
```



```
Epoch 1/800
1/1 ————— 1s 1s/step - loss: 4226.3154
Epoch 2/800
1/1 ————— 0s 48ms/step - loss: 4218.7422
Epoch 3/800
1/1 ————— 0s 41ms/step - loss: 4211.1885
Epoch 4/800
1/1 ————— 0s 39ms/step - loss: 4203.6548
Epoch 5/800
1/1 ————— 0s 40ms/step - loss: 4196.1421
Epoch 6/800
1/1 ————— 0s 82ms/step - loss: 4188.6494
Epoch 7/800
1/1 ————— 0s 41ms/step - loss: 4181.1777
Epoch 8/800
1/1 ————— 0s 39ms/step - loss: 4173.7266
Epoch 9/800
1/1 ————— 0s 38ms/step - loss: 4166.2974
Epoch 10/800
1/1 ————— 0s 39ms/step - loss: 4158.8887
Epoch 11/800
1/1 ————— 0s 38ms/step - loss: 4151.5015
Epoch 12/800
1/1 ————— 0s 40ms/step - loss: 4144.1348
Epoch 13/800
1/1 ————— 0s 39ms/step - loss: 4136.7900
Epoch 14/800
1/1 ————— 0s 38ms/step - loss: 4129.4692
Epoch 15/800
1/1 ————— 0s 39ms/step - loss: 4122.1709
Epoch 16/800
1/1 ————— 0s 40ms/step - loss: 4114.8936
Epoch 17/800
1/1 ————— 0s 40ms/step - loss: 4107.6372
Epoch 18/800
1/1 ————— 0s 41ms/step - loss: 4100.4053
Epoch 19/800
1/1 ————— 0s 43ms/step - loss: 4093.1965
Epoch 20/800
1/1 ————— 0s 48ms/step - loss: 4086.0132
Epoch 21/800
```

1/1 ————— 0s 58ms/step - loss: 4078.8582
Epoch 22/800
1/1 ————— 0s 45ms/step - loss: 4071.7437
Epoch 23/800
1/1 ————— 0s 43ms/step - loss: 4064.6890
Epoch 24/800
1/1 ————— 0s 46ms/step - loss: 4057.7222
Epoch 25/800
1/1 ————— 0s 40ms/step - loss: 4050.7769
Epoch 26/800
1/1 ————— 0s 40ms/step - loss: 4043.8479
Epoch 27/800
1/1 ————— 0s 39ms/step - loss: 4036.9346
Epoch 28/800
1/1 ————— 0s 39ms/step - loss: 4030.0359
Epoch 29/800
1/1 ————— 0s 40ms/step - loss: 4023.1516
Epoch 30/800
1/1 ————— 0s 40ms/step - loss: 4016.2810
Epoch 31/800
1/1 ————— 0s 51ms/step - loss: 4009.4236
Epoch 32/800
1/1 ————— 0s 57ms/step - loss: 4002.5811
Epoch 33/800
1/1 ————— 0s 47ms/step - loss: 3995.7515
Epoch 34/800
1/1 ————— 0s 43ms/step - loss: 3988.9331
Epoch 35/800
1/1 ————— 0s 44ms/step - loss: 3982.1252
Epoch 36/800
1/1 ————— 0s 44ms/step - loss: 3975.3274
Epoch 37/800
1/1 ————— 0s 43ms/step - loss: 3968.5391
Epoch 38/800
1/1 ————— 0s 40ms/step - loss: 3961.7588
Epoch 39/800
1/1 ————— 0s 46ms/step - loss: 3954.9866
Epoch 40/800
1/1 ————— 0s 42ms/step - loss: 3948.2212
Epoch 41/800
1/1 ————— 0s 43ms/step - loss: 3941.4617
Epoch 42/800

```
Epoch 42/800
1/1 ————— 0s 44ms/step - loss: 3934.7104
Epoch 43/800
1/1 ————— 0s 50ms/step - loss: 3927.9690
Epoch 44/800
1/1 ————— 0s 43ms/step - loss: 3921.2322
Epoch 45/800
1/1 ————— 0s 44ms/step - loss: 3914.4990
Epoch 46/800
1/1 ————— 0s 39ms/step - loss: 3907.7688
Epoch 47/800
1/1 ————— 0s 41ms/step - loss: 3901.0410
Epoch 48/800
1/1 ————— 0s 94ms/step - loss: 3894.3223
Epoch 49/800
1/1 ————— 0s 48ms/step - loss: 3887.6055
Epoch 50/800
1/1 ————— 0s 44ms/step - loss: 3880.8896
Epoch 51/800
1/1 ————— 0s 40ms/step - loss: 3874.1804
Epoch 52/800
1/1 ————— 0s 44ms/step - loss: 3867.4763
Epoch 53/800
1/1 ————— 0s 41ms/step - loss: 3860.7751
Epoch 54/800
1/1 ————— 0s 40ms/step - loss: 3854.0850
Epoch 55/800
1/1 ————— 0s 40ms/step - loss: 3847.4041
Epoch 56/800
1/1 ————— 0s 38ms/step - loss: 3840.7356
Epoch 57/800
1/1 ————— 0s 39ms/step - loss: 3834.0889
Epoch 58/800
1/1 ————— 0s 38ms/step - loss: 3827.4590
Epoch 59/800
1/1 ————— 0s 40ms/step - loss: 3820.8230
Epoch 60/800
1/1 ————— 0s 42ms/step - loss: 3814.1809
Epoch 61/800
1/1 ————— 0s 40ms/step - loss: 3807.5317
Epoch 62/800
1/1 ————— 0s 44ms/step - loss: 3800.8750
```

Epoch 63/800
1/1  0s 43ms/step - loss: 3794.2100
Epoch 64/800
1/1  0s 40ms/step - loss: 3787.5361
Epoch 65/800
1/1  0s 47ms/step - loss: 3780.8569
Epoch 66/800
1/1  0s 49ms/step - loss: 3774.1689
Epoch 67/800
1/1  0s 56ms/step - loss: 3767.4712
Epoch 68/800
1/1  0s 65ms/step - loss: 3760.7632
Epoch 69/800
1/1  0s 49ms/step - loss: 3754.0447
Epoch 70/800
1/1  0s 39ms/step - loss: 3747.3193
Epoch 71/800
1/1  0s 41ms/step - loss: 3740.5835
Epoch 72/800
1/1  0s 41ms/step - loss: 3733.8352
Epoch 73/800
1/1  0s 42ms/step - loss: 3727.0793
Epoch 74/800
1/1  0s 39ms/step - loss: 3720.3140
Epoch 75/800
1/1  0s 43ms/step - loss: 3713.5371
Epoch 76/800
1/1  0s 40ms/step - loss: 3706.7559
Epoch 77/800
1/1  0s 41ms/step - loss: 3699.9617
Epoch 78/800
1/1  0s 39ms/step - loss: 3693.1638
Epoch 79/800
1/1  0s 42ms/step - loss: 3686.3550
Epoch 80/800
1/1  0s 41ms/step - loss: 3679.5388
Epoch 81/800
1/1  0s 53ms/step - loss: 3672.7068
Epoch 82/800
1/1  0s 54ms/step - loss: 3665.8584
Epoch 83/800

1/1 ————— 0s 50ms/step - loss: 3658.9937
Epoch 84/800
1/1 ————— 0s 90ms/step - loss: 3652.1111
Epoch 85/800
1/1 ————— 0s 41ms/step - loss: 3645.2117
Epoch 86/800
1/1 ————— 0s 41ms/step - loss: 3638.2944
Epoch 87/800
1/1 ————— 0s 38ms/step - loss: 3631.3596
Epoch 88/800
1/1 ————— 0s 41ms/step - loss: 3624.4062
Epoch 89/800
1/1 ————— 0s 43ms/step - loss: 3617.4351
Epoch 90/800
1/1 ————— 0s 54ms/step - loss: 3610.4451
Epoch 91/800
1/1 ————— 0s 51ms/step - loss: 3603.4360
Epoch 92/800
1/1 ————— 0s 42ms/step - loss: 3596.4087
Epoch 93/800
1/1 ————— 0s 44ms/step - loss: 3589.3613
Epoch 94/800
1/1 ————— 0s 48ms/step - loss: 3582.2954
Epoch 95/800
1/1 ————— 0s 42ms/step - loss: 3575.2090
Epoch 96/800
1/1 ————— 0s 39ms/step - loss: 3568.1038
Epoch 97/800
1/1 ————— 0s 40ms/step - loss: 3560.9780
Epoch 98/800
1/1 ————— 0s 40ms/step - loss: 3553.8325
Epoch 99/800
1/1 ————— 0s 44ms/step - loss: 3546.6685
Epoch 100/800
1/1 ————— 0s 54ms/step - loss: 3539.4841
Epoch 101/800
1/1 ————— 0s 55ms/step - loss: 3532.2793
Epoch 102/800
1/1 ————— 0s 57ms/step - loss: 3525.0537
Epoch 103/800
1/1 ————— 0s 53ms/step - loss: 3517.8074
Epoch 104/800

Epoch 105/800
1/1 ————— 0s 49ms/step - loss: 3510.5400
Epoch 106/800
1/1 ————— 0s 45ms/step - loss: 3503.2520
Epoch 107/800
1/1 ————— 0s 42ms/step - loss: 3495.9421
Epoch 108/800
1/1 ————— 0s 45ms/step - loss: 3488.6118
Epoch 109/800
1/1 ————— 0s 47ms/step - loss: 3481.2610
Epoch 110/800
1/1 ————— 0s 47ms/step - loss: 3473.8887
Epoch 111/800
1/1 ————— 0s 46ms/step - loss: 3466.4941
Epoch 112/800
1/1 ————— 0s 46ms/step - loss: 3459.0784
Epoch 113/800
1/1 ————— 0s 43ms/step - loss: 3451.6404
Epoch 114/800
1/1 ————— 0s 40ms/step - loss: 3444.1807
Epoch 115/800
1/1 ————— 0s 41ms/step - loss: 3436.6987
Epoch 116/800
1/1 ————— 0s 41ms/step - loss: 3429.1943
Epoch 117/800
1/1 ————— 0s 45ms/step - loss: 3421.6680
Epoch 118/800
1/1 ————— 0s 49ms/step - loss: 3414.1191
Epoch 119/800
1/1 ————— 0s 48ms/step - loss: 3406.5479
Epoch 120/800
1/1 ————— 0s 46ms/step - loss: 3398.9541
Epoch 121/800
1/1 ————— 0s 44ms/step - loss: 3391.3374
Epoch 122/800
1/1 ————— 0s 42ms/step - loss: 3383.6985
Epoch 123/800
1/1 ————— 0s 42ms/step - loss: 3376.0371
Epoch 124/800
1/1 ————— 0s 40ms/step - loss: 3368.3523
Epoch 125/800
1/1 ————— 0s 41ms/step - loss: 3360.6453

```
Epoch 125/800
1/1 ————— 0s 39ms/step - loss: 3352.9167
Epoch 126/800
1/1 ————— 0s 40ms/step - loss: 3345.1655
Epoch 127/800
1/1 ————— 0s 44ms/step - loss: 3337.3926
Epoch 128/800
1/1 ————— 0s 40ms/step - loss: 3329.5967
Epoch 129/800
1/1 ————— 0s 51ms/step - loss: 3321.7783
Epoch 130/800
1/1 ————— 0s 46ms/step - loss: 3313.9370
Epoch 131/800
1/1 ————— 0s 46ms/step - loss: 3306.0728
Epoch 132/800
1/1 ————— 0s 47ms/step - loss: 3298.1855
Epoch 133/800
1/1 ————— 0s 41ms/step - loss: 3290.2754
Epoch 134/800
1/1 ————— 0s 44ms/step - loss: 3282.3418
Epoch 135/800
1/1 ————— 0s 49ms/step - loss: 3274.3853
Epoch 136/800
1/1 ————— 0s 41ms/step - loss: 3266.4065
Epoch 137/800
1/1 ————— 0s 40ms/step - loss: 3258.4062
Epoch 138/800
1/1 ————— 0s 42ms/step - loss: 3250.3828
Epoch 139/800
1/1 ————— 0s 47ms/step - loss: 3242.3376
Epoch 140/800
1/1 ————— 0s 44ms/step - loss: 3234.2732
Epoch 141/800
1/1 ————— 0s 43ms/step - loss: 3226.1855
Epoch 142/800
1/1 ————— 0s 41ms/step - loss: 3218.0752
Epoch 143/800
1/1 ————— 0s 45ms/step - loss: 3209.9421
Epoch 144/800
1/1 ————— 0s 40ms/step - loss: 3201.7861
Epoch 145/800
1/1 ————— 0s 41ms/step - loss: 3193.6101
```

```
1/1 _____ 0s 41ms/step - loss: 3135.0101
Epoch 146/800
1/1 _____ 0s 39ms/step - loss: 3185.4109
Epoch 147/800
1/1 _____ 0s 40ms/step - loss: 3177.1890
Epoch 148/800
1/1 _____ 0s 40ms/step - loss: 3168.9446
Epoch 149/800
1/1 _____ 0s 39ms/step - loss: 3160.6770
Epoch 150/800
1/1 _____ 0s 40ms/step - loss: 3152.3860
Epoch 151/800
1/1 _____ 0s 39ms/step - loss: 3144.0742
Epoch 152/800
1/1 _____ 0s 41ms/step - loss: 3135.7412
Epoch 153/800
1/1 _____ 0s 39ms/step - loss: 3127.3860
Epoch 154/800
1/1 _____ 0s 38ms/step - loss: 3119.0081
Epoch 155/800
1/1 _____ 0s 39ms/step - loss: 3110.6072
Epoch 156/800
1/1 _____ 0s 40ms/step - loss: 3102.1836
Epoch 157/800
1/1 _____ 0s 39ms/step - loss: 3093.7400
Epoch 158/800
1/1 _____ 0s 39ms/step - loss: 3085.2744
Epoch 159/800
1/1 _____ 0s 40ms/step - loss: 3076.7861
Epoch 160/800
1/1 _____ 0s 38ms/step - loss: 3068.2754
Epoch 161/800
1/1 _____ 0s 51ms/step - loss: 3059.7432
Epoch 162/800
1/1 _____ 0s 60ms/step - loss: 3051.1943
Epoch 163/800
1/1 _____ 0s 41ms/step - loss: 3042.6240
Epoch 164/800
1/1 _____ 0s 39ms/step - loss: 3034.0317
Epoch 165/800
1/1 _____ 0s 39ms/step - loss: 3025.4170
Epoch 166/800
```

1/1 ————— 0s 41ms/step - loss: 3016.7798
Epoch 167/800
1/1 ————— 0s 40ms/step - loss: 3008.1201
Epoch 168/800
1/1 ————— 0s 39ms/step - loss: 2999.4385
Epoch 169/800
1/1 ————— 0s 39ms/step - loss: 2990.7351
Epoch 170/800
1/1 ————— 0s 43ms/step - loss: 2982.0093
Epoch 171/800
1/1 ————— 0s 44ms/step - loss: 2973.2617
Epoch 172/800
1/1 ————— 0s 39ms/step - loss: 2964.4922
Epoch 173/800
1/1 ————— 0s 38ms/step - loss: 2955.7007
Epoch 174/800
1/1 ————— 0s 41ms/step - loss: 2946.8872
Epoch 175/800
1/1 ————— 0s 38ms/step - loss: 2938.0518
Epoch 176/800
1/1 ————— 0s 40ms/step - loss: 2929.1970
Epoch 177/800
1/1 ————— 0s 42ms/step - loss: 2920.3228
Epoch 178/800
1/1 ————— 0s 40ms/step - loss: 2911.4265
Epoch 179/800
1/1 ————— 0s 38ms/step - loss: 2902.5093
Epoch 180/800
1/1 ————— 0s 40ms/step - loss: 2893.5708
Epoch 181/800
1/1 ————— 0s 40ms/step - loss: 2884.6113
Epoch 182/800
1/1 ————— 0s 41ms/step - loss: 2875.6306
Epoch 183/800
1/1 ————— 0s 42ms/step - loss: 2866.6289
Epoch 184/800
1/1 ————— 0s 39ms/step - loss: 2857.6060
Epoch 185/800
1/1 ————— 0s 39ms/step - loss: 2848.5625
Epoch 186/800
1/1 ————— 0s 40ms/step - loss: 2839.4988
Epoch 187/800

```
Epoch 187/800
1/1 _____ 0s 44ms/step - loss: 2830.4192
Epoch 188/800
1/1 _____ 0s 45ms/step - loss: 2821.3188
Epoch 189/800
1/1 _____ 0s 45ms/step - loss: 2812.1987
Epoch 190/800
1/1 _____ 0s 51ms/step - loss: 2803.0581
Epoch 191/800
1/1 _____ 0s 44ms/step - loss: 2793.8979
Epoch 192/800
1/1 _____ 0s 38ms/step - loss: 2784.7183
Epoch 193/800
1/1 _____ 0s 57ms/step - loss: 2775.5186
Epoch 194/800
1/1 _____ 0s 57ms/step - loss: 2766.2993
Epoch 195/800
1/1 _____ 0s 54ms/step - loss: 2757.0642
Epoch 196/800
1/1 _____ 0s 58ms/step - loss: 2747.8110
Epoch 197/800
1/1 _____ 0s 53ms/step - loss: 2738.5386
Epoch 198/800
1/1 _____ 0s 85ms/step - loss: 2729.2471
Epoch 199/800
1/1 _____ 0s 46ms/step - loss: 2719.9363
Epoch 200/800
1/1 _____ 0s 48ms/step - loss: 2710.6074
Epoch 201/800
1/1 _____ 0s 43ms/step - loss: 2701.2615
Epoch 202/800
1/1 _____ 0s 39ms/step - loss: 2691.9009
Epoch 203/800
1/1 _____ 0s 39ms/step - loss: 2682.5239
Epoch 204/800
1/1 _____ 0s 41ms/step - loss: 2673.1289
Epoch 205/800
1/1 _____ 0s 42ms/step - loss: 2663.7161
Epoch 206/800
1/1 _____ 0s 44ms/step - loss: 2654.2852
Epoch 207/800
1/1 _____ 0s 38ms/step - loss: 2644.8362
```

Epoch 208/800
1/1 ————— 0s 40ms/step - loss: 2635.3696
Epoch 209/800
1/1 ————— 0s 45ms/step - loss: 2625.8870
Epoch 210/800
1/1 ————— 0s 43ms/step - loss: 2616.3901
Epoch 211/800
1/1 ————— 0s 42ms/step - loss: 2606.8755
Epoch 212/800
1/1 ————— 0s 40ms/step - loss: 2597.3438
Epoch 213/800
1/1 ————— 0s 38ms/step - loss: 2587.7944
Epoch 214/800
1/1 ————— 0s 45ms/step - loss: 2578.2285
Epoch 215/800
1/1 ————— 0s 53ms/step - loss: 2568.6455
Epoch 216/800
1/1 ————— 0s 40ms/step - loss: 2559.0459
Epoch 217/800
1/1 ————— 0s 46ms/step - loss: 2549.4302
Epoch 218/800
1/1 ————— 0s 44ms/step - loss: 2539.7981
Epoch 219/800
1/1 ————— 0s 40ms/step - loss: 2530.1499
Epoch 220/800
1/1 ————— 0s 41ms/step - loss: 2520.4858
Epoch 221/800
1/1 ————— 0s 39ms/step - loss: 2510.8066
Epoch 222/800
1/1 ————— 0s 38ms/step - loss: 2501.1116
Epoch 223/800
1/1 ————— 0s 41ms/step - loss: 2491.4014
Epoch 224/800
1/1 ————— 0s 40ms/step - loss: 2481.6765
Epoch 225/800
1/1 ————— 0s 39ms/step - loss: 2471.9368
Epoch 226/800
1/1 ————— 0s 41ms/step - loss: 2462.1826
Epoch 227/800
1/1 ————— 0s 44ms/step - loss: 2452.4141
Epoch 228/800

1/1 ————— 0s 39ms/step - loss: 2442.6316
Epoch 229/800
1/1 ————— 0s 41ms/step - loss: 2432.8354
Epoch 230/800
1/1 ————— 0s 39ms/step - loss: 2423.0256
Epoch 231/800
1/1 ————— 0s 40ms/step - loss: 2413.2026
Epoch 232/800
1/1 ————— 0s 77ms/step - loss: 2403.3667
Epoch 233/800
1/1 ————— 0s 40ms/step - loss: 2393.5178
Epoch 234/800
1/1 ————— 0s 42ms/step - loss: 2383.6567
Epoch 235/800
1/1 ————— 0s 41ms/step - loss: 2373.7832
Epoch 236/800
1/1 ————— 0s 41ms/step - loss: 2363.9014
Epoch 237/800
1/1 ————— 0s 47ms/step - loss: 2354.0076
Epoch 238/800
1/1 ————— 0s 40ms/step - loss: 2344.1025
Epoch 239/800
1/1 ————— 0s 40ms/step - loss: 2334.1868
Epoch 240/800
1/1 ————— 0s 39ms/step - loss: 2324.2603
Epoch 241/800
1/1 ————— 0s 42ms/step - loss: 2314.3232
Epoch 242/800
1/1 ————— 0s 41ms/step - loss: 2304.3760
Epoch 243/800
1/1 ————— 0s 44ms/step - loss: 2294.4204
Epoch 244/800
1/1 ————— 0s 40ms/step - loss: 2284.4551
Epoch 245/800
1/1 ————— 0s 43ms/step - loss: 2274.4800
Epoch 246/800
1/1 ————— 0s 44ms/step - loss: 2264.4966
Epoch 247/800
1/1 ————— 0s 50ms/step - loss: 2254.5042
Epoch 248/800
1/1 ————— 0s 47ms/step - loss: 2244.5034
Epoch 249/800

Epoch 249/800
1/1 ————— 0s 49ms/step - loss: 2234.4946
Epoch 250/800
1/1 ————— 0s 43ms/step - loss: 2224.4780
Epoch 251/800
1/1 ————— 0s 39ms/step - loss: 2214.4536
Epoch 252/800
1/1 ————— 0s 40ms/step - loss: 2204.4224
Epoch 253/800
1/1 ————— 0s 40ms/step - loss: 2194.3838
Epoch 254/800
1/1 ————— 0s 41ms/step - loss: 2184.3394
Epoch 255/800
1/1 ————— 0s 44ms/step - loss: 2174.2881
Epoch 256/800
1/1 ————— 0s 46ms/step - loss: 2164.2312
Epoch 257/800
1/1 ————— 0s 40ms/step - loss: 2154.1689
Epoch 258/800
1/1 ————— 0s 41ms/step - loss: 2144.1011
Epoch 259/800
1/1 ————— 0s 43ms/step - loss: 2134.0298
Epoch 260/800
1/1 ————— 0s 101ms/step - loss: 2123.9563
Epoch 261/800
1/1 ————— 0s 38ms/step - loss: 2113.8789
Epoch 262/800
1/1 ————— 0s 42ms/step - loss: 2103.7976
Epoch 263/800
1/1 ————— 0s 39ms/step - loss: 2093.7129
Epoch 264/800
1/1 ————— 0s 43ms/step - loss: 2083.6250
Epoch 265/800
1/1 ————— 0s 41ms/step - loss: 2073.5344
Epoch 266/800
1/1 ————— 0s 44ms/step - loss: 2063.4414
Epoch 267/800
1/1 ————— 0s 45ms/step - loss: 2053.3464
Epoch 268/800
1/1 ————— 0s 38ms/step - loss: 2043.2498
Epoch 269/800
1/1 ————— 0s 40ms/step - loss: 2033.1514

Epoch 270/800
1/1 ————— 0s 40ms/step - loss: 2023.0522
Epoch 271/800
1/1 ————— 0s 47ms/step - loss: 2012.9521
Epoch 272/800
1/1 ————— 0s 43ms/step - loss: 2002.8521
Epoch 273/800
1/1 ————— 0s 41ms/step - loss: 1992.7515
Epoch 274/800
1/1 ————— 0s 45ms/step - loss: 1982.6514
Epoch 275/800
1/1 ————— 0s 41ms/step - loss: 1972.5520
Epoch 276/800
1/1 ————— 0s 45ms/step - loss: 1962.4536
Epoch 277/800
1/1 ————— 0s 46ms/step - loss: 1952.3599
Epoch 278/800
1/1 ————— 0s 44ms/step - loss: 1942.2686
Epoch 279/800
1/1 ————— 0s 38ms/step - loss: 1932.1799
Epoch 280/800
1/1 ————— 0s 40ms/step - loss: 1922.0936
Epoch 281/800
1/1 ————— 0s 46ms/step - loss: 1912.0105
Epoch 282/800
1/1 ————— 0s 52ms/step - loss: 1901.9309
Epoch 283/800
1/1 ————— 0s 48ms/step - loss: 1891.8546
Epoch 284/800
1/1 ————— 0s 41ms/step - loss: 1881.7827
Epoch 285/800
1/1 ————— 0s 41ms/step - loss: 1871.7148
Epoch 286/800
1/1 ————— 0s 44ms/step - loss: 1861.6521
Epoch 287/800
1/1 ————— 0s 44ms/step - loss: 1851.5941
Epoch 288/800
1/1 ————— 0s 39ms/step - loss: 1841.5416
Epoch 289/800
1/1 ————— 0s 39ms/step - loss: 1831.4951
Epoch 290/800
1/1 ————— 0s 32ms/step - loss: 1821.4546

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1/1 ————— 0s 33ms/step - loss: 1821.4340
Epoch 291/800
1/1 ————— 0s 40ms/step - loss: 1811.4216
Epoch 292/800
1/1 ————— 0s 39ms/step - loss: 1801.3987
Epoch 293/800
1/1 ————— 0s 41ms/step - loss: 1791.3831
Epoch 294/800
1/1 ————— 0s 41ms/step - loss: 1781.3755
Epoch 295/800
1/1 ————— 0s 40ms/step - loss: 1771.3757
Epoch 296/800
1/1 ————— 0s 39ms/step - loss: 1761.3845
Epoch 297/800
1/1 ————— 0s 40ms/step - loss: 1751.4031
Epoch 298/800
1/1 ————— 0s 38ms/step - loss: 1741.4309
Epoch 299/800
1/1 ————— 0s 41ms/step - loss: 1731.4685
Epoch 300/800
1/1 ————— 0s 38ms/step - loss: 1721.5161
Epoch 301/800
1/1 ————— 0s 43ms/step - loss: 1711.5739
Epoch 302/800
1/1 ————— 0s 38ms/step - loss: 1701.6423
Epoch 303/800
1/1 ————— 0s 40ms/step - loss: 1691.7219
Epoch 304/800
1/1 ————— 0s 38ms/step - loss: 1681.8126
Epoch 305/800
1/1 ————— 0s 40ms/step - loss: 1671.9175
Epoch 306/800
1/1 ————— 0s 39ms/step - loss: 1662.0356
Epoch 307/800
1/1 ————— 0s 39ms/step - loss: 1652.1665
Epoch 308/800
1/1 ————— 0s 39ms/step - loss: 1642.3102
Epoch 309/800
1/1 ————— 0s 39ms/step - loss: 1632.4669
Epoch 310/800
1/1 ————— 0s 41ms/step - loss: 1622.6372
Epoch 311/800
```

1/1 ————— 0s 39ms/step - loss: 1612.8213
Epoch 312/800
1/1 ————— 0s 40ms/step - loss: 1603.0199
Epoch 313/800
1/1 ————— 0s 38ms/step - loss: 1593.2329
Epoch 314/800
1/1 ————— 0s 39ms/step - loss: 1583.4607
Epoch 315/800
1/1 ————— 0s 46ms/step - loss: 1573.7041
Epoch 316/800
1/1 ————— 0s 75ms/step - loss: 1563.9630
Epoch 317/800
1/1 ————— 0s 137ms/step - loss: 1554.2378
Epoch 318/800
1/1 ————— 0s 122ms/step - loss: 1544.5288
Epoch 319/800
1/1 ————— 0s 57ms/step - loss: 1534.8367
Epoch 320/800
1/1 ————— 0s 52ms/step - loss: 1525.1614
Epoch 321/800
1/1 ————— 0s 53ms/step - loss: 1515.5059
Epoch 322/800
1/1 ————— 0s 58ms/step - loss: 1505.8682
Epoch 323/800
1/1 ————— 0s 50ms/step - loss: 1496.2485
Epoch 324/800
1/1 ————— 0s 48ms/step - loss: 1486.6473
Epoch 325/800
1/1 ————— 0s 50ms/step - loss: 1477.0647
Epoch 326/800
1/1 ————— 0s 52ms/step - loss: 1467.5012
Epoch 327/800
1/1 ————— 0s 52ms/step - loss: 1457.9572
Epoch 328/800
1/1 ————— 0s 69ms/step - loss: 1448.4329
Epoch 329/800
1/1 ————— 0s 62ms/step - loss: 1438.9285
Epoch 330/800
1/1 ————— 0s 64ms/step - loss: 1429.4446
Epoch 331/800
1/1 ————— 0s 65ms/step - loss: 1419.9814

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Epoch 332/800
1/1 ————— 0s 57ms/step - loss: 1410.5394
Epoch 333/800
1/1 ————— 0s 55ms/step - loss: 1401.1189
Epoch 334/800
1/1 ————— 0s 56ms/step - loss: 1391.7200
Epoch 335/800
1/1 ————— 0s 48ms/step - loss: 1382.3433
Epoch 336/800
1/1 ————— 0s 72ms/step - loss: 1372.9889
Epoch 337/800
1/1 ————— 0s 60ms/step - loss: 1363.6573
Epoch 338/800
1/1 ————— 0s 61ms/step - loss: 1354.3497
Epoch 339/800
1/1 ————— 0s 57ms/step - loss: 1345.0657
Epoch 340/800
1/1 ————— 0s 55ms/step - loss: 1335.8054
Epoch 341/800
1/1 ————— 0s 98ms/step - loss: 1326.5691
Epoch 342/800
1/1 ————— 0s 57ms/step - loss: 1317.3574
Epoch 343/800
1/1 ————— 0s 46ms/step - loss: 1308.1704
Epoch 344/800
1/1 ————— 0s 48ms/step - loss: 1299.0085
Epoch 345/800
1/1 ————— 0s 48ms/step - loss: 1289.8722
Epoch 346/800
1/1 ————— 0s 42ms/step - loss: 1280.7614
Epoch 347/800
1/1 ————— 0s 44ms/step - loss: 1271.6765
Epoch 348/800
1/1 ————— 0s 50ms/step - loss: 1262.6179
Epoch 349/800
1/1 ————— 0s 45ms/step - loss: 1253.5857
Epoch 350/800
1/1 ————— 0s 48ms/step - loss: 1244.5806
Epoch 351/800
1/1 ————— 0s 55ms/step - loss: 1235.6025
Epoch 352/800
1/1 ————— 0s 47ms/step - loss: 1226.6517
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Epoch 353/800
1/1 ————— 0s 49ms/step - loss: 1217.7288
Epoch 354/800
1/1 ————— 0s 47ms/step - loss: 1208.8339
Epoch 355/800
1/1 ————— 0s 53ms/step - loss: 1199.9672
Epoch 356/800
1/1 ————— 0s 43ms/step - loss: 1191.1292
Epoch 357/800
1/1 ————— 0s 69ms/step - loss: 1182.3198
Epoch 358/800
1/1 ————— 0s 62ms/step - loss: 1173.5397
Epoch 359/800
1/1 ————— 0s 61ms/step - loss: 1164.7891
Epoch 360/800
1/1 ————— 0s 107ms/step - loss: 1156.0681
Epoch 361/800
1/1 ————— 0s 76ms/step - loss: 1147.3772
Epoch 362/800
1/1 ————— 0s 51ms/step - loss: 1138.7163
Epoch 363/800
1/1 ————— 0s 58ms/step - loss: 1130.0862
Epoch 364/800
1/1 ————— 0s 60ms/step - loss: 1121.4866
Epoch 365/800
1/1 ————— 0s 159ms/step - loss: 1112.9182
Epoch 366/800
1/1 ————— 0s 46ms/step - loss: 1104.3809
Epoch 367/800
1/1 ————— 0s 49ms/step - loss: 1095.8752
Epoch 368/800
1/1 ————— 0s 46ms/step - loss: 1087.4014
Epoch 369/800
1/1 ————— 0s 43ms/step - loss: 1078.9596
Epoch 370/800
1/1 ————— 0s 51ms/step - loss: 1070.5498
Epoch 371/800
1/1 ————— 0s 47ms/step - loss: 1062.1730
Epoch 372/800
1/1 ————— 0s 46ms/step - loss: 1053.8295
Epoch 373/800
```

1/1 ————— 0s 59ms/step - loss: 1045.5192
Epoch 374/800
1/1 ————— 0s 53ms/step - loss: 1037.2422
Epoch 375/800
1/1 ————— 0s 57ms/step - loss: 1028.9988
Epoch 376/800
1/1 ————— 0s 57ms/step - loss: 1020.7889
Epoch 377/800
1/1 ————— 0s 56ms/step - loss: 1012.6130
Epoch 378/800
1/1 ————— 0s 45ms/step - loss: 1004.4715
Epoch 379/800
1/1 ————— 0s 43ms/step - loss: 996.3642
Epoch 380/800
1/1 ————— 0s 45ms/step - loss: 988.2916
Epoch 381/800
1/1 ————— 0s 52ms/step - loss: 980.2538
Epoch 382/800
1/1 ————— 0s 45ms/step - loss: 972.2510
Epoch 383/800
1/1 ————— 0s 51ms/step - loss: 964.2833
Epoch 384/800
1/1 ————— 0s 45ms/step - loss: 956.3512
Epoch 385/800
1/1 ————— 0s 50ms/step - loss: 948.4546
Epoch 386/800
1/1 ————— 0s 47ms/step - loss: 940.5939
Epoch 387/800
1/1 ————— 0s 48ms/step - loss: 932.7691
Epoch 388/800
1/1 ————— 0s 93ms/step - loss: 924.9806
Epoch 389/800
1/1 ————— 0s 42ms/step - loss: 917.2283
Epoch 390/800
1/1 ————— 0s 41ms/step - loss: 909.5127
Epoch 391/800
1/1 ————— 0s 42ms/step - loss: 901.8339
Epoch 392/800
1/1 ————— 0s 47ms/step - loss: 894.1918
Epoch 393/800
1/1 ————— 0s 48ms/step - loss: 886.5868
Epoch 394/800

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Epoch 394/800
1/1 ————— 0s 40ms/step - loss: 879.0190
Epoch 395/800
1/1 ————— 0s 47ms/step - loss: 871.4887
Epoch 396/800
1/1 ————— 0s 41ms/step - loss: 863.9960
Epoch 397/800
1/1 ————— 0s 43ms/step - loss: 856.5410
Epoch 398/800
1/1 ————— 0s 41ms/step - loss: 849.1239
Epoch 399/800
1/1 ————— 0s 38ms/step - loss: 841.7448
Epoch 400/800
1/1 ————— 0s 41ms/step - loss: 834.4038
Epoch 401/800
1/1 ————— 0s 49ms/step - loss: 827.1012
Epoch 402/800
1/1 ————— 0s 43ms/step - loss: 819.8369
Epoch 403/800
1/1 ————— 0s 41ms/step - loss: 812.6115
Epoch 404/800
1/1 ————— 0s 40ms/step - loss: 805.4252
Epoch 405/800
1/1 ————— 0s 43ms/step - loss: 798.2780
Epoch 406/800
1/1 ————— 0s 41ms/step - loss: 791.1696
Epoch 407/800
1/1 ————— 0s 41ms/step - loss: 784.1005
Epoch 408/800
1/1 ————— 0s 43ms/step - loss: 777.0705
Epoch 409/800
1/1 ————— 0s 40ms/step - loss: 770.0798
Epoch 410/800
1/1 ————— 0s 42ms/step - loss: 763.1288
Epoch 411/800
1/1 ————— 0s 40ms/step - loss: 756.2172
Epoch 412/800
1/1 ————— 0s 47ms/step - loss: 749.3453
Epoch 413/800
1/1 ————— 0s 44ms/step - loss: 742.5131
Epoch 414/800
1/1 ————— 0s 63ms/step - loss: 735.7208
```



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Epoch 415/800
1/1 ————— 0s 40ms/step - loss: 728.9684
Epoch 416/800
1/1 ————— 0s 39ms/step - loss: 722.2561
Epoch 417/800
1/1 ————— 0s 38ms/step - loss: 715.5838
Epoch 418/800
1/1 ————— 0s 39ms/step - loss: 708.9518
Epoch 419/800
1/1 ————— 0s 41ms/step - loss: 702.3599
Epoch 420/800
1/1 ————— 0s 40ms/step - loss: 695.8085
Epoch 421/800
1/1 ————— 0s 41ms/step - loss: 689.2975
Epoch 422/800
1/1 ————— 0s 40ms/step - loss: 682.8269
Epoch 423/800
1/1 ————— 0s 41ms/step - loss: 676.3967
Epoch 424/800
1/1 ————— 0s 39ms/step - loss: 670.0074
Epoch 425/800
1/1 ————— 0s 75ms/step - loss: 663.6585
Epoch 426/800
1/1 ————— 0s 39ms/step - loss: 657.3504
Epoch 427/800
1/1 ————— 0s 42ms/step - loss: 651.0831
Epoch 428/800
1/1 ————— 0s 40ms/step - loss: 644.8565
Epoch 429/800
1/1 ————— 0s 44ms/step - loss: 638.6708
Epoch 430/800
1/1 ————— 0s 46ms/step - loss: 632.5259
Epoch 431/800
1/1 ————— 0s 42ms/step - loss: 626.4221
Epoch 432/800
1/1 ————— 0s 40ms/step - loss: 620.3590
Epoch 433/800
1/1 ————— 0s 43ms/step - loss: 614.3369
Epoch 434/800
1/1 ————— 0s 40ms/step - loss: 608.3557
Epoch 435/800
1/1 ————— 0s 44ms/step - loss: 602.4155
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1/1 ————— 0s 44ms/step - loss: 602.4155
Epoch 436/800
1/1 ————— 0s 82ms/step - loss: 596.5165
Epoch 437/800
1/1 ————— 0s 38ms/step - loss: 590.6586
Epoch 438/800
1/1 ————— 0s 40ms/step - loss: 584.8415
Epoch 439/800
1/1 ————— 0s 41ms/step - loss: 579.0656
Epoch 440/800
1/1 ————— 0s 41ms/step - loss: 573.3311
Epoch 441/800
1/1 ————— 0s 40ms/step - loss: 567.6376
Epoch 442/800
1/1 ————— 0s 41ms/step - loss: 561.9852
Epoch 443/800
1/1 ————— 0s 42ms/step - loss: 556.3738
Epoch 444/800
1/1 ————— 0s 40ms/step - loss: 550.8034
Epoch 445/800
1/1 ————— 0s 41ms/step - loss: 545.2739
Epoch 446/800
1/1 ————— 0s 39ms/step - loss: 539.7855
Epoch 447/800
1/1 ————— 0s 39ms/step - loss: 534.3379
Epoch 448/800
1/1 ————— 0s 41ms/step - loss: 528.9313
Epoch 449/800
1/1 ————— 0s 39ms/step - loss: 523.5656
Epoch 450/800
1/1 ————— 0s 40ms/step - loss: 518.2406
Epoch 451/800
1/1 ————— 0s 38ms/step - loss: 512.9564
Epoch 452/800
1/1 ————— 0s 80ms/step - loss: 507.7130
Epoch 453/800
1/1 ————— 0s 43ms/step - loss: 502.5104
Epoch 454/800
1/1 ————— 0s 40ms/step - loss: 497.3484
Epoch 455/800
1/1 ————— 0s 42ms/step - loss: 492.2270
Epoch 456/800

1/1 ————— 0s 41ms/step - loss: 487.1462
Epoch 457/800
1/1 ————— 0s 39ms/step - loss: 482.1058
Epoch 458/800
1/1 ————— 0s 42ms/step - loss: 477.1060
Epoch 459/800
1/1 ————— 0s 46ms/step - loss: 472.1464
Epoch 460/800
1/1 ————— 0s 38ms/step - loss: 467.2272
Epoch 461/800
1/1 ————— 0s 42ms/step - loss: 462.3481
Epoch 462/800
1/1 ————— 0s 42ms/step - loss: 457.5093
Epoch 463/800
1/1 ————— 0s 47ms/step - loss: 452.7105
Epoch 464/800
1/1 ————— 0s 42ms/step - loss: 447.9517
Epoch 465/800
1/1 ————— 0s 46ms/step - loss: 443.2328
Epoch 466/800
1/1 ————— 0s 43ms/step - loss: 438.5537
Epoch 467/800
1/1 ————— 0s 43ms/step - loss: 433.9145
Epoch 468/800
1/1 ————— 0s 41ms/step - loss: 429.3147
Epoch 469/800
1/1 ————— 0s 43ms/step - loss: 424.7546
Epoch 470/800
1/1 ————— 0s 93ms/step - loss: 420.2338
Epoch 471/800
1/1 ————— 0s 45ms/step - loss: 415.7525
Epoch 472/800
1/1 ————— 0s 43ms/step - loss: 411.3104
Epoch 473/800
1/1 ————— 0s 42ms/step - loss: 406.9073
Epoch 474/800
1/1 ————— 0s 43ms/step - loss: 402.5433
Epoch 475/800
1/1 ————— 0s 48ms/step - loss: 398.2182
Epoch 476/800
1/1 ————— 0s 39ms/step - loss: 393.9319

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Epoch 477/800
1/1 _____ 0s 43ms/step - loss: 389.6842
Epoch 478/800
1/1 _____ 0s 44ms/step - loss: 385.4753
Epoch 479/800
1/1 _____ 0s 39ms/step - loss: 381.3045
Epoch 480/800
1/1 _____ 0s 41ms/step - loss: 377.1721
Epoch 481/800
1/1 _____ 0s 40ms/step - loss: 373.0778
Epoch 482/800
1/1 _____ 0s 45ms/step - loss: 369.0216
Epoch 483/800
1/1 _____ 0s 45ms/step - loss: 365.0033
Epoch 484/800
1/1 _____ 0s 42ms/step - loss: 361.0228
Epoch 485/800
1/1 _____ 0s 39ms/step - loss: 357.0798
Epoch 486/800
1/1 _____ 0s 42ms/step - loss: 353.1743
Epoch 487/800
1/1 _____ 0s 40ms/step - loss: 349.3062
Epoch 488/800
1/1 _____ 0s 42ms/step - loss: 345.4750
Epoch 489/800
1/1 _____ 0s 41ms/step - loss: 341.6809
Epoch 490/800
1/1 _____ 0s 44ms/step - loss: 337.9236
Epoch 491/800
1/1 _____ 0s 92ms/step - loss: 334.2029
Epoch 492/800
1/1 _____ 0s 42ms/step - loss: 330.5189
Epoch 493/800
1/1 _____ 0s 40ms/step - loss: 326.8712
Epoch 494/800
1/1 _____ 0s 44ms/step - loss: 323.2599
Epoch 495/800
1/1 _____ 0s 45ms/step - loss: 319.6845
Epoch 496/800
1/1 _____ 0s 52ms/step - loss: 316.1450
Epoch 497/800
1/1 _____ 0s 41ms/step - loss: 312.6414
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Epoch 498/800
1/1 _____ 0s 39ms/step - loss: 309.1732
Epoch 499/800
1/1 _____ 0s 43ms/step - loss: 305.7406
Epoch 500/800
1/1 _____ 0s 44ms/step - loss: 302.3432
Epoch 501/800
1/1 _____ 0s 41ms/step - loss: 298.9809
Epoch 502/800
1/1 _____ 0s 39ms/step - loss: 295.6534
Epoch 503/800
1/1 _____ 0s 41ms/step - loss: 292.3606
Epoch 504/800
1/1 _____ 0s 39ms/step - loss: 289.1023
Epoch 505/800
1/1 _____ 0s 39ms/step - loss: 285.8784
Epoch 506/800
1/1 _____ 0s 38ms/step - loss: 282.6883
Epoch 507/800
1/1 _____ 0s 48ms/step - loss: 279.5323
Epoch 508/800
1/1 _____ 0s 43ms/step - loss: 276.4100
Epoch 509/800
1/1 _____ 0s 43ms/step - loss: 273.3214
Epoch 510/800
1/1 _____ 0s 41ms/step - loss: 270.2660
Epoch 511/800
1/1 _____ 0s 41ms/step - loss: 267.2439
Epoch 512/800
1/1 _____ 0s 86ms/step - loss: 264.2547
Epoch 513/800
1/1 _____ 0s 40ms/step - loss: 261.2984
Epoch 514/800
1/1 _____ 0s 41ms/step - loss: 258.3747
Epoch 515/800
1/1 _____ 0s 41ms/step - loss: 255.4834
Epoch 516/800
1/1 _____ 0s 46ms/step - loss: 252.6242
Epoch 517/800
1/1 _____ 0s 39ms/step - loss: 249.7971
Epoch 518/800
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1/1 ————— 0s 38ms/step - loss: 247.0018
Epoch 519/800
1/1 ————— 0s 43ms/step - loss: 244.2381
Epoch 520/800
1/1 ————— 0s 45ms/step - loss: 241.5058
Epoch 521/800
1/1 ————— 0s 42ms/step - loss: 238.8047
Epoch 522/800
1/1 ————— 0s 45ms/step - loss: 236.1345
Epoch 523/800
1/1 ————— 0s 44ms/step - loss: 233.4952
Epoch 524/800
1/1 ————— 0s 42ms/step - loss: 230.8864
Epoch 525/800
1/1 ————— 0s 44ms/step - loss: 228.3080
Epoch 526/800
1/1 ————— 0s 40ms/step - loss: 225.7597
Epoch 527/800
1/1 ————— 0s 42ms/step - loss: 223.2414
Epoch 528/800
1/1 ————— 0s 43ms/step - loss: 220.7528
Epoch 529/800
1/1 ————— 0s 44ms/step - loss: 218.2938
Epoch 530/800
1/1 ————— 0s 44ms/step - loss: 215.8640
Epoch 531/800
1/1 ————— 0s 41ms/step - loss: 213.4633
Epoch 532/800
1/1 ————— 0s 90ms/step - loss: 211.0915
Epoch 533/800
1/1 ————— 0s 40ms/step - loss: 208.7484
Epoch 534/800
1/1 ————— 0s 42ms/step - loss: 206.4337
Epoch 535/800
1/1 ————— 0s 42ms/step - loss: 204.1471
Epoch 536/800
1/1 ————— 0s 42ms/step - loss: 201.8887
Epoch 537/800
1/1 ————— 0s 40ms/step - loss: 199.6580
Epoch 538/800
1/1 ————— 0s 43ms/step - loss: 197.4549
Epoch 539/800

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Epoch 539/800
1/1 _____ 0s 39ms/step - loss: 195.2791
Epoch 540/800
1/1 _____ 0s 43ms/step - loss: 193.1305
Epoch 541/800
1/1 _____ 0s 40ms/step - loss: 191.0087
Epoch 542/800
1/1 _____ 0s 39ms/step - loss: 188.9137
Epoch 543/800
1/1 _____ 0s 45ms/step - loss: 186.8452
Epoch 544/800
1/1 _____ 0s 41ms/step - loss: 184.8030
Epoch 545/800
1/1 _____ 0s 39ms/step - loss: 182.7867
Epoch 546/800
1/1 _____ 0s 41ms/step - loss: 180.7963
Epoch 547/800
1/1 _____ 0s 44ms/step - loss: 178.8315
Epoch 548/800
1/1 _____ 0s 40ms/step - loss: 176.8920
Epoch 549/800
1/1 _____ 0s 42ms/step - loss: 174.9777
Epoch 550/800
1/1 _____ 0s 40ms/step - loss: 173.0883
Epoch 551/800
1/1 _____ 0s 41ms/step - loss: 171.2237
Epoch 552/800
1/1 _____ 0s 81ms/step - loss: 169.3835
Epoch 553/800
1/1 _____ 0s 38ms/step - loss: 167.5676
Epoch 554/800
1/1 _____ 0s 39ms/step - loss: 165.7758
Epoch 555/800
1/1 _____ 0s 40ms/step - loss: 164.0078
Epoch 556/800
1/1 _____ 0s 42ms/step - loss: 162.2634
Epoch 557/800
1/1 _____ 0s 37ms/step - loss: 160.5424
Epoch 558/800
1/1 _____ 0s 44ms/step - loss: 158.8446
Epoch 559/800
1/1 _____ 0s 40ms/step - loss: 157.1697
```

Epoch 560/800
1/1 ————— 0s 41ms/step - loss: 155.5175
Epoch 561/800
1/1 ————— 0s 39ms/step - loss: 153.8880
Epoch 562/800
1/1 ————— 0s 41ms/step - loss: 152.2806
Epoch 563/800
1/1 ————— 0s 39ms/step - loss: 150.6953
Epoch 564/800
1/1 ————— 0s 43ms/step - loss: 149.1320
Epoch 565/800
1/1 ————— 0s 45ms/step - loss: 147.5903
Epoch 566/800
1/1 ————— 0s 40ms/step - loss: 146.0699
Epoch 567/800
1/1 ————— 0s 43ms/step - loss: 144.5708
Epoch 568/800
1/1 ————— 0s 40ms/step - loss: 143.0927
Epoch 569/800
1/1 ————— 0s 46ms/step - loss: 141.6354
Epoch 570/800
1/1 ————— 0s 38ms/step - loss: 140.1987
Epoch 571/800
1/1 ————— 0s 85ms/step - loss: 138.7823
Epoch 572/800
1/1 ————— 0s 43ms/step - loss: 137.3861
Epoch 573/800
1/1 ————— 0s 41ms/step - loss: 136.0098
Epoch 574/800
1/1 ————— 0s 45ms/step - loss: 134.6532
Epoch 575/800
1/1 ————— 0s 39ms/step - loss: 133.3161
Epoch 576/800
1/1 ————— 0s 38ms/step - loss: 131.9984
Epoch 577/800
1/1 ————— 0s 41ms/step - loss: 130.6998
Epoch 578/800
1/1 ————— 0s 39ms/step - loss: 129.4200
Epoch 579/800
1/1 ————— 0s 41ms/step - loss: 128.1589
Epoch 580/800

1/1 ————— 0s 38ms/step - loss: 126.9164
Epoch 581/800
1/1 ————— 0s 41ms/step - loss: 125.6920
Epoch 582/800
1/1 ————— 0s 41ms/step - loss: 124.4857
Epoch 583/800
1/1 ————— 0s 38ms/step - loss: 123.2973
Epoch 584/800
1/1 ————— 0s 40ms/step - loss: 122.1266
Epoch 585/800
1/1 ————— 0s 40ms/step - loss: 120.9733
Epoch 586/800
1/1 ————— 0s 40ms/step - loss: 119.8373
Epoch 587/800
1/1 ————— 0s 38ms/step - loss: 118.7184
Epoch 588/800
1/1 ————— 0s 38ms/step - loss: 117.6163
Epoch 589/800
1/1 ————— 0s 99ms/step - loss: 116.5308
Epoch 590/800
1/1 ————— 0s 45ms/step - loss: 115.4619
Epoch 591/800
1/1 ————— 0s 41ms/step - loss: 114.4092
Epoch 592/800
1/1 ————— 0s 43ms/step - loss: 113.3725
Epoch 593/800
1/1 ————— 0s 50ms/step - loss: 112.3518
Epoch 594/800
1/1 ————— 0s 49ms/step - loss: 111.3468
Epoch 595/800
1/1 ————— 0s 41ms/step - loss: 110.3573
Epoch 596/800
1/1 ————— 0s 42ms/step - loss: 109.3831
Epoch 597/800
1/1 ————— 0s 42ms/step - loss: 108.4241
Epoch 598/800
1/1 ————— 0s 41ms/step - loss: 107.4799
Epoch 599/800
1/1 ————— 0s 42ms/step - loss: 106.5506
Epoch 600/800
1/1 ————— 0s 40ms/step - loss: 105.6358
Epoch 601/800

```
1/1 ————— 0s 46ms/step - loss: 104.7354
Epoch 602/800
1/1 ————— 0s 39ms/step - loss: 103.8492
Epoch 603/800
1/1 ————— 0s 41ms/step - loss: 102.9770
Epoch 604/800
1/1 ————— 0s 83ms/step - loss: 102.1187
Epoch 605/800
1/1 ————— 0s 41ms/step - loss: 101.2741
Epoch 606/800
1/1 ————— 0s 44ms/step - loss: 100.4429
Epoch 607/800
1/1 ————— 0s 41ms/step - loss: 99.6251
Epoch 608/800
1/1 ————— 0s 39ms/step - loss: 98.8204
Epoch 609/800
1/1 ————— 0s 42ms/step - loss: 98.0287
Epoch 610/800
1/1 ————— 0s 47ms/step - loss: 97.2498
Epoch 611/800
1/1 ————— 0s 40ms/step - loss: 96.4835
Epoch 612/800
1/1 ————— 0s 40ms/step - loss: 95.7297
Epoch 613/800
1/1 ————— 0s 39ms/step - loss: 94.9882
Epoch 614/800
1/1 ————— 0s 41ms/step - loss: 94.2588
Epoch 615/800
1/1 ————— 0s 38ms/step - loss: 93.5414
Epoch 616/800
1/1 ————— 0s 41ms/step - loss: 92.8358
Epoch 617/800
1/1 ————— 0s 39ms/step - loss: 92.1418
Epoch 618/800
1/1 ————— 0s 63ms/step - loss: 91.4593
Epoch 619/800
1/1 ————— 0s 60ms/step - loss: 90.7882
Epoch 620/800
1/1 ————— 0s 38ms/step - loss: 90.1282
Epoch 621/800
1/1 ————— 0s 38ms/step - loss: 89.4792
```

```
Epoch 622/800
1/1 ————— 0s 40ms/step - loss: 88.8410
Epoch 623/800
1/1 ————— 0s 38ms/step - loss: 88.2136
Epoch 624/800
1/1 ————— 0s 41ms/step - loss: 87.5967
Epoch 625/800
1/1 ————— 0s 38ms/step - loss: 86.9902
Epoch 626/800
1/1 ————— 0s 39ms/step - loss: 86.3940
Epoch 627/800
1/1 ————— 0s 39ms/step - loss: 85.8078
Epoch 628/800
1/1 ————— 0s 40ms/step - loss: 85.2317
Epoch 629/800
1/1 ————— 0s 40ms/step - loss: 84.6653
Epoch 630/800
1/1 ————— 0s 44ms/step - loss: 84.1086
Epoch 631/800
1/1 ————— 0s 39ms/step - loss: 83.5614
Epoch 632/800
1/1 ————— 0s 39ms/step - loss: 83.0237
Epoch 633/800
1/1 ————— 0s 41ms/step - loss: 82.4952
Epoch 634/800
1/1 ————— 0s 38ms/step - loss: 81.9758
Epoch 635/800
1/1 ————— 0s 42ms/step - loss: 81.4653
Epoch 636/800
1/1 ————— 0s 39ms/step - loss: 80.9637
Epoch 637/800
1/1 ————— 0s 39ms/step - loss: 80.4708
Epoch 638/800
1/1 ————— 0s 43ms/step - loss: 79.9865
Epoch 639/800
1/1 ————— 0s 40ms/step - loss: 79.5106
Epoch 640/800
1/1 ————— 0s 44ms/step - loss: 79.0431
Epoch 641/800
1/1 ————— 0s 43ms/step - loss: 78.5838
Epoch 642/800
1/1 ————— 0s 40ms/step - loss: 78.1325
```

```
1/1 _____ 0s 40ms/step - loss: 70.1325
Epoch 643/800
1/1 _____ 0s 102ms/step - loss: 77.6891
Epoch 644/800
1/1 _____ 0s 45ms/step - loss: 77.2536
Epoch 645/800
1/1 _____ 0s 45ms/step - loss: 76.8257
Epoch 646/800
1/1 _____ 0s 47ms/step - loss: 76.4055
Epoch 647/800
1/1 _____ 0s 53ms/step - loss: 75.9926
Epoch 648/800
1/1 _____ 0s 43ms/step - loss: 75.5872
Epoch 649/800
1/1 _____ 0s 43ms/step - loss: 75.1890
Epoch 650/800
1/1 _____ 0s 44ms/step - loss: 74.7978
Epoch 651/800
1/1 _____ 0s 44ms/step - loss: 74.4136
Epoch 652/800
1/1 _____ 0s 43ms/step - loss: 74.0364
Epoch 653/800
1/1 _____ 0s 42ms/step - loss: 73.6659
Epoch 654/800
1/1 _____ 0s 39ms/step - loss: 73.3020
Epoch 655/800
1/1 _____ 0s 49ms/step - loss: 72.9447
Epoch 656/800
1/1 _____ 0s 42ms/step - loss: 72.5939
Epoch 657/800
1/1 _____ 0s 41ms/step - loss: 72.2494
Epoch 658/800
1/1 _____ 0s 38ms/step - loss: 71.9111
Epoch 659/800
1/1 _____ 0s 42ms/step - loss: 71.5790
Epoch 660/800
1/1 _____ 0s 100ms/step - loss: 71.2529
Epoch 661/800
1/1 _____ 0s 44ms/step - loss: 70.9327
Epoch 662/800
1/1 _____ 0s 47ms/step - loss: 70.6184
Epoch 663/800
```

1/1 ————— 0s 49ms/step - loss: 70.3097
Epoch 664/800
1/1 ————— 0s 48ms/step - loss: 70.0068
Epoch 665/800
1/1 ————— 0s 42ms/step - loss: 69.7094
Epoch 666/800
1/1 ————— 0s 46ms/step - loss: 69.4174
Epoch 667/800
1/1 ————— 0s 46ms/step - loss: 69.1308
Epoch 668/800
1/1 ————— 0s 40ms/step - loss: 68.8494
Epoch 669/800
1/1 ————— 0s 39ms/step - loss: 68.5732
Epoch 670/800
1/1 ————— 0s 42ms/step - loss: 68.3022
Epoch 671/800
1/1 ————— 0s 86ms/step - loss: 68.0360
Epoch 672/800
1/1 ————— 0s 50ms/step - loss: 67.7749
Epoch 673/800
1/1 ————— 0s 59ms/step - loss: 67.5185
Epoch 674/800
1/1 ————— 0s 56ms/step - loss: 67.2669
Epoch 675/800
1/1 ————— 0s 57ms/step - loss: 67.0199
Epoch 676/800
1/1 ————— 0s 51ms/step - loss: 66.7776
Epoch 677/800
1/1 ————— 0s 50ms/step - loss: 66.5397
Epoch 678/800
1/1 ————— 0s 59ms/step - loss: 66.3062
Epoch 679/800
1/1 ————— 0s 49ms/step - loss: 66.0771
Epoch 680/800
1/1 ————— 0s 45ms/step - loss: 65.8523
Epoch 681/800
1/1 ————— 0s 48ms/step - loss: 65.6316
Epoch 682/800
1/1 ————— 0s 49ms/step - loss: 65.4150
Epoch 683/800
1/1 ————— 0s 50ms/step - loss: 65.2025
Epoch 684/800

```
Epoch 684/800
1/1 ————— 0s 53ms/step - loss: 64.9940
Epoch 685/800
1/1 ————— 0s 109ms/step - loss: 64.7893
Epoch 686/800
1/1 ————— 0s 55ms/step - loss: 64.5885
Epoch 687/800
1/1 ————— 0s 54ms/step - loss: 64.3914
Epoch 688/800
1/1 ————— 0s 51ms/step - loss: 64.1980
Epoch 689/800
1/1 ————— 0s 48ms/step - loss: 64.0082
Epoch 690/800
1/1 ————— 0s 45ms/step - loss: 63.8220
Epoch 691/800
1/1 ————— 0s 48ms/step - loss: 63.6392
Epoch 692/800
1/1 ————— 0s 55ms/step - loss: 63.4599
Epoch 693/800
1/1 ————— 0s 48ms/step - loss: 63.2839
Epoch 694/800
1/1 ————— 0s 47ms/step - loss: 63.1112
Epoch 695/800
1/1 ————— 0s 50ms/step - loss: 62.9418
Epoch 696/800
1/1 ————— 0s 54ms/step - loss: 62.7755
Epoch 697/800
1/1 ————— 0s 57ms/step - loss: 62.6124
Epoch 698/800
1/1 ————— 0s 47ms/step - loss: 62.4522
Epoch 699/800
1/1 ————— 0s 45ms/step - loss: 62.2951
Epoch 700/800
1/1 ————— 0s 49ms/step - loss: 62.1410
Epoch 701/800
1/1 ————— 0s 46ms/step - loss: 61.9897
Epoch 702/800
1/1 ————— 0s 49ms/step - loss: 61.8412
Epoch 703/800
1/1 ————— 0s 42ms/step - loss: 61.6955
Epoch 704/800
1/1 ————— 0s 43ms/step - loss: 61.5526
```

Epoch 705/800
1/1 ————— 0s 47ms/step - loss: 61.4123
Epoch 706/800
1/1 ————— 0s 43ms/step - loss: 61.2746
Epoch 707/800
1/1 ————— 0s 44ms/step - loss: 61.1395
Epoch 708/800
1/1 ————— 0s 38ms/step - loss: 61.0070
Epoch 709/800
1/1 ————— 0s 87ms/step - loss: 60.8768
Epoch 710/800
1/1 ————— 0s 41ms/step - loss: 60.7491
Epoch 711/800
1/1 ————— 0s 40ms/step - loss: 60.6238
Epoch 712/800
1/1 ————— 0s 48ms/step - loss: 60.5008
Epoch 713/800
1/1 ————— 0s 42ms/step - loss: 60.3800
Epoch 714/800
1/1 ————— 0s 41ms/step - loss: 60.2616
Epoch 715/800
1/1 ————— 0s 40ms/step - loss: 60.1453
Epoch 716/800
1/1 ————— 0s 43ms/step - loss: 60.0311
Epoch 717/800
1/1 ————— 0s 39ms/step - loss: 59.9191
Epoch 718/800
1/1 ————— 0s 38ms/step - loss: 59.8091
Epoch 719/800
1/1 ————— 0s 40ms/step - loss: 59.7012
Epoch 720/800
1/1 ————— 0s 81ms/step - loss: 59.5952
Epoch 721/800
1/1 ————— 0s 42ms/step - loss: 59.4911
Epoch 722/800
1/1 ————— 0s 40ms/step - loss: 59.3890
Epoch 723/800
1/1 ————— 0s 40ms/step - loss: 59.2887
Epoch 724/800
1/1 ————— 0s 45ms/step - loss: 59.1903
Epoch 725/800

1/1  0s 37ms/step - loss: 59.0936
Epoch 726/800
1/1  0s 43ms/step - loss: 58.9987
Epoch 727/800
1/1  0s 40ms/step - loss: 58.9055
Epoch 728/800
1/1  0s 39ms/step - loss: 58.8140
Epoch 729/800
1/1  0s 39ms/step - loss: 58.7242
Epoch 730/800
1/1  0s 42ms/step - loss: 58.6359
Epoch 731/800
1/1  0s 39ms/step - loss: 58.5493
Epoch 732/800
1/1  0s 40ms/step - loss: 58.4641
Epoch 733/800
1/1  0s 40ms/step - loss: 58.3805
Epoch 734/800
1/1  0s 40ms/step - loss: 58.2984
Epoch 735/800
1/1  0s 48ms/step - loss: 58.2177
Epoch 736/800
1/1  0s 51ms/step - loss: 58.1385
Epoch 737/800
1/1  0s 53ms/step - loss: 58.0606
Epoch 738/800
1/1  0s 71ms/step - loss: 57.9841
Epoch 739/800
1/1  0s 41ms/step - loss: 57.9090
Epoch 740/800
1/1  0s 38ms/step - loss: 57.8351
Epoch 741/800
1/1  0s 37ms/step - loss: 57.7626
Epoch 742/800
1/1  0s 40ms/step - loss: 57.6912
Epoch 743/800
1/1  0s 40ms/step - loss: 57.6211
Epoch 744/800
1/1  0s 40ms/step - loss: 57.5522
Epoch 745/800
1/1  0s 38ms/step - loss: 57.4845
Epoch 746/800

Epoch 747/800

1/1 ————— 0s 42ms/step - loss: 57.4179

Epoch 748/800

1/1 ————— 0s 42ms/step - loss: 57.3524

Epoch 749/800

1/1 ————— 0s 49ms/step - loss: 57.2881

Epoch 750/800

1/1 ————— 0s 78ms/step - loss: 57.2248

Epoch 751/800

1/1 ————— 0s 44ms/step - loss: 57.1626

Epoch 752/800

1/1 ————— 0s 43ms/step - loss: 57.1014

Epoch 753/800

1/1 ————— 0s 42ms/step - loss: 57.0412

Epoch 754/800

1/1 ————— 0s 41ms/step - loss: 56.9819

Epoch 755/800

1/1 ————— 0s 40ms/step - loss: 56.9237

Epoch 756/800

1/1 ————— 0s 39ms/step - loss: 56.8664

Epoch 757/800

1/1 ————— 0s 41ms/step - loss: 56.8100

Epoch 758/800

1/1 ————— 0s 48ms/step - loss: 56.7545

Epoch 759/800

1/1 ————— 0s 40ms/step - loss: 56.6999

Epoch 760/800

1/1 ————— 0s 39ms/step - loss: 56.6461

Epoch 761/800

1/1 ————— 0s 39ms/step - loss: 56.5932

Epoch 762/800

1/1 ————— 0s 41ms/step - loss: 56.5412

Epoch 763/800

1/1 ————— 0s 52ms/step - loss: 56.4899

Epoch 764/800

1/1 ————— 0s 61ms/step - loss: 56.4394

Epoch 765/800

1/1 ————— 0s 41ms/step - loss: 56.3897

Epoch 766/800

1/1 ————— 0s 41ms/step - loss: 56.3407

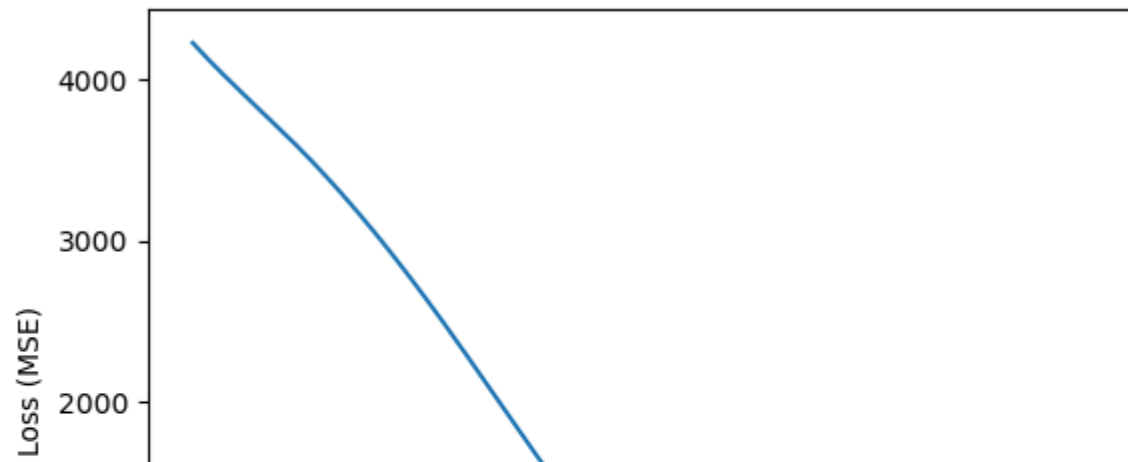
Epoch 767/800

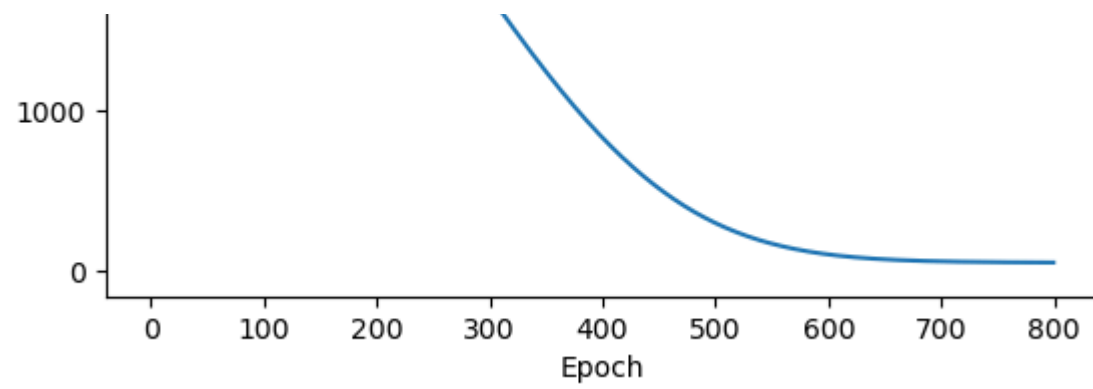
1/1 ————— 0s 40ms/step - loss: 56.2925

```
Epoch 767/800
1/1 ————— 0s 37ms/step - loss: 56.2450
Epoch 768/800
1/1 ————— 0s 42ms/step - loss: 56.1982
Epoch 769/800
1/1 ————— 0s 41ms/step - loss: 56.1520
Epoch 770/800
1/1 ————— 0s 45ms/step - loss: 56.1066
Epoch 771/800
1/1 ————— 0s 40ms/step - loss: 56.0618
Epoch 772/800
1/1 ————— 0s 40ms/step - loss: 56.0176
Epoch 773/800
1/1 ————— 0s 85ms/step - loss: 55.9741
Epoch 774/800
1/1 ————— 0s 40ms/step - loss: 55.9312
Epoch 775/800
1/1 ————— 0s 46ms/step - loss: 55.8889
Epoch 776/800
1/1 ————— 0s 47ms/step - loss: 55.8472
Epoch 777/800
1/1 ————— 0s 49ms/step - loss: 55.8060
Epoch 778/800
1/1 ————— 0s 69ms/step - loss: 55.7654
Epoch 779/800
1/1 ————— 0s 71ms/step - loss: 55.7254
Epoch 780/800
1/1 ————— 0s 55ms/step - loss: 55.6858
Epoch 781/800
1/1 ————— 0s 81ms/step - loss: 55.6468
Epoch 782/800
1/1 ————— 0s 47ms/step - loss: 55.6083
Epoch 783/800
1/1 ————— 0s 47ms/step - loss: 55.5703
Epoch 784/800
1/1 ————— 0s 54ms/step - loss: 55.5328
Epoch 785/800
1/1 ————— 0s 61ms/step - loss: 55.4957
Epoch 786/800
1/1 ————— 0s 49ms/step - loss: 55.4592
Epoch 787/800
1/1 ————— 0s 55ms/step - loss: 55.4220
```

```
1/1 ————— 0s 55ms/step - loss: 55.4230
Epoch 788/800
1/1 ————— 0s 58ms/step - loss: 55.3873
Epoch 789/800
1/1 ————— 0s 52ms/step - loss: 55.3521
Epoch 790/800
1/1 ————— 0s 54ms/step - loss: 55.3173
Epoch 791/800
1/1 ————— 0s 50ms/step - loss: 55.2828
Epoch 792/800
1/1 ————— 0s 41ms/step - loss: 55.2488
Epoch 793/800
1/1 ————— 0s 90ms/step - loss: 55.2152
Epoch 794/800
1/1 ————— 0s 52ms/step - loss: 55.1819
Epoch 795/800
1/1 ————— 0s 77ms/step - loss: 55.1490
Epoch 796/800
1/1 ————— 0s 60ms/step - loss: 55.1165
Epoch 797/800
1/1 ————— 0s 59ms/step - loss: 55.0843
Epoch 798/800
1/1 ————— 0s 85ms/step - loss: 55.0525
Epoch 799/800
1/1 ————— 0s 52ms/step - loss: 55.0210
Epoch 800/800
1/1 ————— 0s 55ms/step - loss: 54.9898
```

Training Loss Over Time





✓ Step 5: Test the Model (Make Predictions)

Give the model new study hours and let it guess the scores.

```
# Predict scores for new study hours
new_hours = np.array([2, 4, 6, 8], dtype=float)
predicted_scores = model.predict(new_hours)

print("Study Hours:", new_hours)
print("Predicted Scores:", predicted_scores.flatten().round(1))
```

```
1/1  0s 74ms/step
Study Hours: [2. 4. 6. 8.]
Predicted Scores: [30.7 51.9 73. 94.2]
```

✓ Step 6: Visualize What the Model Learned

- **Blue dots:** actual data we gave the model
- **Red line:** what the model predicts across the same hours If the red line follows the blue dots, learning happened!

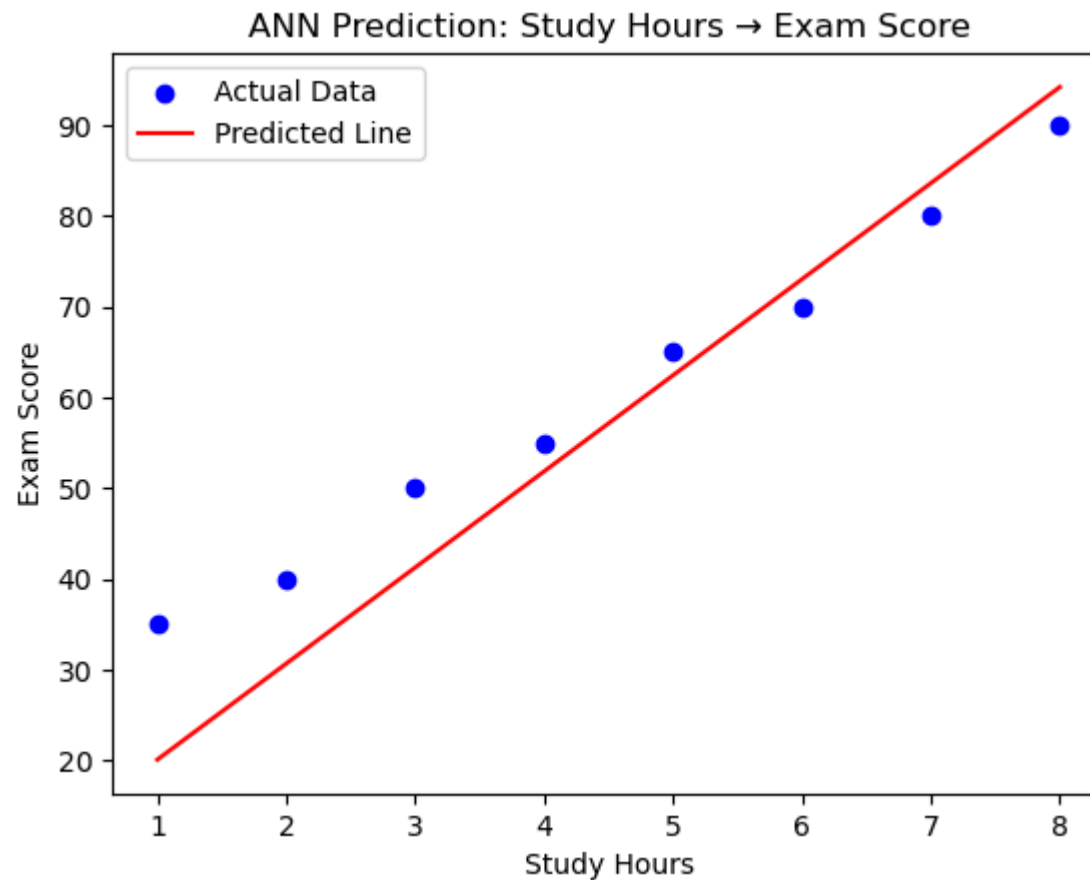
```
# Plot actual data points
plt.scatter(study_hours, exam_scores, color='blue', label='Actual Data')

# Plot model predictions as a red line
line_hours = np.linspace(1, 8, 50).astype(float)
line_preds = model.predict(line_hours)
plt.plot(line_hours, line_preds, color='red', label='Predicted Line')

plt.xlabel('Study Hours')
plt.ylabel('Exam Score')
plt.title('ANN Prediction: Study Hours → Exam Score')
```

```
plt.legend()
plt.show()
```

2/2 ————— 0s 19ms/step



How the ANN Learns — Simple Table

Concept	What It Does	Easy Analogy
Activation Function (ReLU)	Lets useful signals pass, drops negatives	Like highlighting only important lines in a book
Optimizer (Adam)	Improves the model a bit after each round	A teacher giving feedback after every practice test

Concept	What It Does	Easy Analogy
Loss (MSE)	Measures how wrong the predictions are	Like comparing your guessed marks with your real marks

In short: Activation decides what to pass, Loss measures wrongness, Optimizer fixes mistakes.

✓ Lab Task for submission:

1. Predict **your** study hours (e.g., `model.predict(np.array([5.5]))`).
2. Add a new data point (e.g., 9 hours → 95 marks) and retrain.
3. Reduce the **Neurons** from 64 → 32, retrain the model and observe the loss and prediction?
4. Reduce the **Neurons** from 32 → 16, retrain the model and observe the loss and prediction?
5. Reduce the **epochs** from 1000 → 500 or 50. Does the loss go down more?
6. Change `'relu'` to `'sigmoid'` in the hidden layer and compare.